Butterfly gradient variances at half filling and four particles

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Abstract

For the single increasing-stride butterfly with uniform phase and rotation angles, we calculate the first-phase gradient variance of the Walsh-spread magic-block input. At half filling, $k = n/2$ and $n = 4 \cdot 2^r$, an exact five-state transfer gives $V_{ij}^{(k)}(n)/H_k(n) \sim (273/224)(9/8)^r$ for each of the six readouts with $0 \le i < j \le 3$, where H_k is the exact Haar variance. Thus a comparison uniform over all even $2 \le k \le n/2$ cannot have a finite upper constant. A positive lower comparison does hold on this growing family for $n \ge 8$. We also complete the six $k = 4$ readouts, proving their exact formulas, the bounds $H_4/4 < V_{ij}^{(4)} < H_4$, and their leading relative deficit. The half-filling obstruction leaves the sparse regime $n - 2k = \Omega(n)$ open.

1 The scalar and the range of a uniform comparison

We use the input and butterfly architecture of Kerenidis [1], keeping the first phase on mode zero. The exact Haar scalar, rather than a relative-design estimate, is the relevant benchmark for Claim 1, equation (40), of that paper. In correspondence of 18 October [2], Kerenidis suggested a scalar comparison with constants uniform in particle number, reported finite-size evidence up to half filling, and identified the recurrence family of two remaining four-particle readouts. We verify the latter observation and show that half filling requires a separate upper-bound statement. This does not refute the sparse trainability assertion in Theorem 19 of [1], whose conclusion concerns $k \le \gamma_0 n$ for a sufficiently small constant γ_0.

All labels start at zero. Let $n = 2^K$ and let k be even, $2 \le k \le n/2$. Before Walsh spreading the state is

$$|\Psi_k\rangle = \bigotimes_{a=0}^{k/2-1} \frac{|1100\rangle + |0011\rangle}{\sqrt{2}} \otimes |0\rangle^{\otimes(n-2k)}.$$

The normalized Walsh matrix is $H_n(a,c) = (-1)^{\mathrm{popcount}(a\&c)}/\sqrt{n}$. Write $\rho_{n,k}$ for the spread two-particle RDM on $\Lambda^2\mathbb{C}^n$, normalized by $\operatorname{tr}\rho_{n,k} = \binom{k}{2}$. The butterfly is $W_n = R_{K-1}D_{K-1}\cdots R_0 D_0$, with independent uniform angles modulo 2π. Here D_ℓ is diagonal with entries $e^{i\phi_{\ell,a}}$, and R_ℓ acts on pairs differing in bit ℓ by $\left(\begin{smallmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{smallmatrix}\right)$. In particular, $\mathbb{E}\cos^4\theta = 3/8$; the two-mode blocks are not Haar distributed. Set

$$L_{ij} = \langle i \wedge j|\Lambda^2(W_n)\rho_{n,k}\Lambda^2(W_n)^*|i \wedge j\rangle, \qquad V_{ij}^{(k)}(n) = \mathbb{E}(\partial_{\phi_{0,0}} L_{ij})^2.$$

The derivative has mean zero by periodicity. A fixed diagonal data phase can be absorbed into the first uniform phase layer.

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The exact Haar identity of [1, Theorem 9 and Appendix A.2] is

$$H_k(n) = \frac{4\|B\|_F^2 + 2\|\Gamma(B)\|_F^2}{n^2(n^2-1)}, \quad B = [d\Lambda^2(|0\rangle\langle 0|), \rho_{n,k}], \quad \Gamma(B)_{ac} = \sum_d B_{ad,cd}.$$

Pair indices in the contraction are extended antisymmetrically. For this input, direct contraction gives

$$\|B\|_F^2 = \frac{k((k+2)n - 2k^2 + 3k - 6)}{2n^2}, \qquad \|\Gamma(B)\|_F^2 = \frac{k(k-1)^2(n-2k)}{n^2}, \tag{1}$$

$$H_k(n) = \frac{2k((k^2-k+3)(n-2k) + 7k - 6)}{n^4(n^2-1)}. \tag{2}$$

For clarity, the first norm can be obtained before spreading: put $u = n^{-1/2}(1,\dots,1)$, $G = d\Lambda^2(|u\rangle\langle u|)$, and let R be the unspread RDM. Since G is a projector, $\|[G,R]\|_F^2 = 2(\mathrm{tr}(GR^2) - \mathrm{tr}(GRGR))$. The two traces are $k(k+2)/(4n)$ and $k(2k^2-3k+6)/(4n^2)$, respectively. The second norm in (1) follows from $\Gamma(R) = (k-1)P/2$, where P projects onto the $2k$ active modes. The vanishing of this second norm at $k = n/2$ explains why this endpoint must be checked independently of fixed sparse filling.

2 An exact growing family at half filling

Put $\mathcal{A} = \{\{0,1\},\{0,3\},\{1,2\},\{2,3\}\}$ and $\mathcal{C} = \{\{0,2\},\{1,3\}\}$.

Theorem 1. *Let $n = 4\cdot 2^r$, $r \geq 0$, and $k = n/2$. For the six readouts above,*

$$n^4 V_{ij}^{(n/2)}(n) = \begin{cases} \frac{273}{64}(9/8)^r - \frac{7}{2} + \frac{535}{64}(9/16)^r - \frac{57}{8}(1/2)^r, & \{i,j\} \in \mathcal{A}, \\ \frac{273}{64}(9/8)^r - \frac{7}{2} + \frac{161}{64}(9/16)^r - \frac{105}{32}(1/2)^r, & \{i,j\} \in \mathcal{C}. \end{cases} \tag{3}$$

Consequently, for either family,

$$\frac{V_{ij}^{(n/2)}(n)}{H_{n/2}(n)} = \frac{273}{224}(9/8)^r - 1 + O((9/16)^r) \longrightarrow \infty.$$

For every $r \geq 1$ the ratio exceeds $1/4$.

Here is a finite transfer derivation, including the matrices needed to check the formula without fitting values of r. Write $c = 4h + c_0$, $u = 4g + u_0$ and $v = 4(h+g) + v_0$, where high labels lie in $\mathbb{F}_2^r$ and low labels in $\{0,1,2,3\}$; addition of labels means bitwise exclusive-or. Define

$$d_{ab,st} = \chi_a(s)\chi_b(t) - \chi_a(t)\chi_b(s), \quad \chi_a(s) = (-1)^{\mathrm{popcount}(a\&s)},$$

$$C_{ab,uv} = \frac{1}{2}(d_{ab,01} + d_{ab,23})(d_{uv,01} + d_{uv,23}) - \frac{1}{4}\sum_{0\leq s<t<4} d_{ab,st}d_{uv,st}.$$

At half filling the unspread RDM is, with P_a projecting onto the four modes of block a,

$$R = \tfrac{1}{4}I_{\Lambda^2} + \sum_{a=0}^{n/4-1}(|\psi_a\rangle\langle\psi_a| - \tfrac{1}{4}\Lambda^2 P_a), \quad \psi_a = (e_{4a}\wedge e_{4a+1} + e_{4a+2}\wedge e_{4a+3})/\sqrt{2}.$$

Walsh character orthogonality therefore gives, for $u, v \neq 0$,

$$(\rho_{n,n/2})_{0c,uv} = \frac{1}{4n} C_{0c_0,u_0v_0}\, \mathbf{1}_{h+g+v_{\text{high}}=0}. \tag{4}$$

The nonzero C_{0c_0,u_0v_0} are $C_{01,01} = 4$, $C_{01,23} = 8$, $C_{02,02} = -4$, $C_{03,03} = -4$, together with negatives on reversal of the second pair. Thus all nonzero terms have $c_0 + u_0 + v_0 = 0$ and $c_0 \neq 0$. Also $(\rho_{n,n/2})_{0c,cv} = 0$ when $v \neq 0$.

Write $x_a = (W_n)_{ia}$, $y_a = (W_n)_{ja}$ and $z_{ab} = x_a y_b - x_b y_a$. Differentiating D_0 gives $\partial L = i(T - \overline{T})$, with $T = \sum_{c>0,\, 0<u<v} (\rho_{n,n/2})_{0c,uv} z_{0c} \overline{z_{uv}}$. The first-layer phases give distinct frequencies for distinct disjoint triples; the shared-index terms vanish by (4). Consequently

$$V_{ij}^{(n/2)}(n) = \frac{1}{16n^2} \sum_{\substack{c,u,v \neq 0 \text{ distinct} \\ c+u+v=0}} C_{0c_0,u_0v_0}^2\, \mathbb{E}|z_{0c}|^2 |z_{uv}|^2. \tag{5}$$

The sum uses ordered u, v. Its factor accounts for both orientations of a Fourier frequency and for $z_{vu} = -z_{uv}$.

Expand the four wedge factors in the expectation into sixteen terms, indexed by a mask $m \in \{0, \ldots, 15\}$. Their column labels are the four linear forms $0, h, g, h+g$; masks reverse the order in each of $(0,h), (g,h+g), (0,h), (g,h+g)$, the first two being ket factors. When high bits are summed from the most significant bit downwards, equality of the high suffixes is encoded by the kernel of $(a,b) \mapsto ah + bg$. There are exactly five states, in the order

$$\mathbb{F}_2^2, \quad \langle (1,0) \rangle, \quad \langle (0,1) \rangle, \quad \langle (1,1) \rangle, \quad \{0\}.$$

For a new bit choice $f \in (\mathbb{F}_2^2)^*$, the state changes from S to $S \cap \ker f$. Integrating the phases requires equal ket and bra counts in each new coset, separately in each output row. For each old coset and row, if a factors stay and b flip, the angular factor is

$$a(a,b) = \mathbb{E}(\cos^a \theta \sin^b \theta) = \begin{cases} (2p)!(2q)!/(4^{p+q} p! q! (p+q)!), & a = 2p,\ b = 2q, \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

For these high layers the two output rows use distinct gates, since $i, j < 4$ already differ in a low bit. Flip signs disappear in every nonzero angular integral. This gives the transfer independently of r.

After combining masks related by interchange of the middle two line states, (5) becomes

$$n^4 V_{ij}^{(n/2)}(n) = e^{\mathsf{T}} (M_A^r \ell_A + M_B^r \ell_B + M_C^r \ell_C), \qquad e = (1,0,0,0,0)^{\mathsf{T}}, \tag{7}$$

where

$$M_A = \frac{1}{16} \begin{pmatrix} 9 & 1 & 9 & 1 & 0 \\ 0 & 8 & 0 & 0 & 8 \\ 0 & 0 & 18 & 0 & 2 \\ 0 & 0 & 0 & 8 & 8 \\ 0 & 0 & 0 & 0 & 16 \end{pmatrix}, \quad M_B = \frac{1}{16} \begin{pmatrix} 9 & 1 & 0 & 0 & 0 \\ 0 & 8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$M_C = \frac{1}{16} \begin{pmatrix} 9 & 1 & 0 & 1 & 0 \\ 0 & 8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 8 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The terminal vectors, obtained by integrating the two low-bit layers, are

	$\mathcal{A}$	$\mathcal{C}$
ℓ_A	$(9/8, 53/8, 49/64, 49/16, 7/2)^{\mathsf{T}}$	$(3/16, 77/32, 49/64, 361/64, 7/2)^{\mathsf{T}}$
ℓ_B	$(3/4, 15/4, 0, 0, 0)^{\mathsf{T}}$	$(-3/8, 19/16, 0, 0, 0)^{\mathsf{T}}$
ℓ_C	$(1/8, 5/8, 0, 1/16, 0)^{\mathsf{T}}$	$(3/16, 13/32, 0, 41/64, 0)^{\mathsf{T}}$

(8)

For the five states, use the respective low-layer representatives

$$(h, g) = (0, 0), (0, 1), (1, 0), (1, 1), (1, 2).$$

Sum over c_0, u_0, put $v_0 = c_0 + u_0$, weight by C^2, and use (6) along the unique paths through the first two layers. Exclude $u = 0$ when $u_0 = 0$ and $(0, 1) \in S$, and exclude $v = 0$ when $v_0 = 0$ and $(1, 1) \in S$. Mask signs are $(-1)^{\mathrm{popcount}(m)}$. Masks $0, 15$ give M_A before combining with $5, 10$; masks $1, 2, 4, 7, 8, 11, 13, 14$ give M_B; masks $3, 12$ give M_C before combining with $6, 9$. The matrices displayed are four times the moment transfer, because $n^2/16 = 4^r$ in (5). The third coordinate of ℓ_C is unreachable from e and is set to zero.

Completion of the proof of Theorem 1. The transfer derivation proves (7). Direct multiplication gives $P(M_A) = P(M_B) = P(M_C) = 0$ for

$$P(x) = x(x - 1)(x - 1/2)(x - 9/16)(x - 9/8).$$

Thus both sides of (3) obey the same recurrence of order five. Substitution of (8) in (7), for $r = 0, 1, 2, 3, 4$, gives the displayed coefficients and proves the identities for every $r \geq 0$. This is an exact matrix calculation; it does not extrapolate a finite table. Equation (2) gives

$$n^4 H_{n/2}(n) = \frac{n(7n/2 - 6)}{n^2 - 1} = \frac{7}{2} + O(1/n),$$

which proves the ratio asymptotic. Let $F_C(r)$ be the second expression in (3). It is increasing:

$$F_C(r + 1) - F_C(r) = \frac{273}{512}(9/8)^r - \frac{1127}{1024}(9/16)^r + \frac{105}{64}(1/2)^r > 0.$$

For $r \geq 2$ the first term exceeds the magnitude of the second, and $r = 0, 1$ are direct rational checks. Since $F_C(1) = 1099/1024$ and $n^4 H_{n/2} < 7/2$, its ratio for $r \geq 1$ exceeds $1099/3584 > 1/4$. The first expression exceeds $F_C(r)$ by $[187(9/16)^r - 123(1/2)^r]/32 > 0$. □

3 Completion of the four-particle readouts

Proposition 2. *Put $n = 8 \cdot 2^s$, $s \geq 0$. Then*

$$n^5 V_{ij}^{(4)}(n) = \begin{cases} A_4(s), & \{i, j\} \in \mathcal{A}, \\ C_4(s), & \{i, j\} \in \mathcal{C}, \end{cases}$$

where

$$A_4(s) = 120 - \tfrac{891}{8}(3/4)^s + \tfrac{533}{64}(9/16)^s + \tfrac{179}{2}(1/2)^s - \tfrac{911}{8}(3/8)^s + \tfrac{8567}{128}(9/32)^s - 40(1/4)^s,$$
$$C_4(s) = 120 - \tfrac{891}{8}(3/4)^s + \tfrac{533}{64}(9/16)^s - \tfrac{287}{4}(1/2)^s + \tfrac{125}{2}(3/8)^s + \tfrac{5281}{128}(9/32)^s - \tfrac{323}{8}(1/4)^s.$$

For every one of these six readouts,

$$\frac{1}{4} < \frac{V_{ij}^{(4)}(n)}{H_4(n)} < 1, \quad H_4(n) = \frac{8(15n - 98)}{n^4(n^2 - 1)}, \quad 1 - \frac{V_{ij}^{(4)}(n)}{H_4(n)} = \frac{297}{320}(3/4)^s + O((9/16)^s).$$

We give the exact recurrence behind the coefficient calculation. This also specifies the computer-assisted part of this proposition. Set $q = 8$, $m = 2^s$, and $R = \rho_{8,4}$. For rows i, j put $X_a = \sum_{h<m}(W_n)_{i,qh+a}$, $Y_a = \sum_{h<m}(W_n)_{j,qh+a}$, $\alpha = (W_n)_{i0}$ and $\beta = (W_n)_{j0}$. With $z_{ab} = X_aY_b - X_bY_a$ and $d_{ab} = \mathbf{1}_{a=0}(\alpha Y_b - X_b\beta)$ for $a < b$, differentiation gives

$$\partial L = im^{-2}T, \quad T = \sum_{a<b,\, u<v} R_{ab,uv}(d_{ab}\overline{z}_{uv} - z_{ab}\overline{d}_{uv}), \quad V = m^{-4}\mathbb{E}(-T^2).$$

Independent subblocks A, B at the next layer give

$$X' = uX^A + vX^B, \quad Y' = wY^A + zY^B, \quad \alpha' = u\alpha^A, \quad \beta' = w\beta^A,$$

where $u = \cos\theta e^{i\eta_1}$, $v = -\sin\theta e^{i\eta_2}$, $w = \cos\vartheta e^{i\eta_3}$ and $z = -\sin\vartheta e^{i\eta_4}$ are independent between rows, with uniform phases. Order variables as $(X, Y, \overline{X}, \overline{Y}, \alpha, \beta, \overline{\alpha}, \overline{\beta})$ and write $M_s(e) = \mathbb{E}Z^e$. Split the unmarked exponents between f and $e - f$, putting every marked exponent in f. Each split has coefficient

$$\prod_{\nu<4q} \binom{e_\nu}{f_\nu}\, a(d_X(f), d_X(e-f))a(d_Y(f), d_Y(e-f)),$$

and contributes that coefficient times $M_s(f)M_s(e-f)$. Here d_X, d_Y count all factors in their row, including conjugates and marks. Keep a split only if ket and bra degrees balance in each row of each branch, and in each component column of each branch, counting the marks as column zero. These rules follow from the independent new phases and from simultaneous shifts of first-layer phases in each residue class. Initial moments are unique-path phase and angle integrals in W_8.

Closing the surviving monomials of $-T^2$ under these splits gives 2,988 moments and 10,003 transitions. The supplied rational certificate gives every exponent, transition, base moment and exponential-polynomial solution for each readout. The verifier checks each initial value and the entire coefficient identity $M_{s+1}(e) = \sum_f a_{e,f}M_s(f)M_s(e-f)$, rather than a finite list of layers. Its scalar contraction gives A_4 for $01, 03, 12, 23$ and C_4 for $02, 13$. This proves the asserted all-layer formulas, including Kerenidis's two suggested pairs $12, 23$. The same certificate includes the preceding $q = 4, k = 2$ calculation: for $n = 4 \cdot 2^t$, all five pairs $01, 02, 12, 13, 23$ have relative deficit $(17/20)(3/4)^t + O((9/16)^t)$. This is a fixed-particle effect, distinct from Theorem 1.

For completeness, the bounds need no extrapolation. If $h_s = n^5H_4(n)$, then $120 - 98\,2^{-s} \le h_s < 120$. We have $A_4(s) > C_4(s)$: their difference is

$$\tfrac{645}{4}2^{-s} - \tfrac{1411}{8}(3/8)^s + \tfrac{1643}{64}(9/32)^s + \tfrac{3}{8}(1/4)^s > 0.$$

For $s \ge 2$, dropping positive terms gives $C_4(s) \ge 120 - \frac{891}{8}(3/4)^s - \frac{287}{4}2^{-s} - \frac{323}{8}4^{-s} \ge 2361/64 > 30$. For $s \ge 3$,

$$h_s - A_4(s) \ge (3/4)^s\left[\tfrac{891}{8} - \tfrac{375}{2}(2/3)^s - \tfrac{533}{64}(3/4)^s - \tfrac{8567}{128}(3/8)^s\right] > 0;$$

the bracket increases and is positive at $s = 3$. Direct rational substitution for $s = 0, 1, 2$ completes the bounds. The leading deficit follows by dividing the formulas by h_s.

4 Exact finite-size comparison and the remaining question

The tables in the verification archive cover every even $k \le n/2$ at $n = 16, 32, 64, 128$ for the representative retaining pair 01 and disjoint pair 13. All entries are exact fractions. Table 1 reports the extrema, rounded only for display; it makes no claim about other readout geometries.

n	pair	minimum V/H (at k)	maximum V/H (at k)
16	01	0.729873 (4)	0.930905 (2)
16	13	0.336871 (2)	0.597442 (8)
32	01	0.588903 (8)	0.956252 (16)
32	13	0.511095 (8)	0.787483 (16)
64	01	0.536444 (16)	1.093098 (32)
64	13	0.518345 (16)	0.991897 (32)
128	01	0.516801 (32)	1.284232 (64)
128	13	0.512561 (32)	1.223721 (64)

Table 1: Exact enumeration over the stated particle numbers, displayed as decimals.

The computation uses a first-layer Fourier reduction independent of the layer recurrence. For general $b = k/2$, define $S_b(a) = \sum_{t<b} \chi_a(t)$. The spread matrix entries needed by the derivative are

$$\begin{aligned} n^2(\rho_{n,k})_{0c,uv} = {} & C_{0c_0,u_0v_0} S_b((c+u+v)_{\text{high}}) \\ & + 4\mathbf{1}_{u_0=0,c_0=v_0} S_b(u_{\text{high}}) S_b((c+v)_{\text{high}}) \\ & - 4\mathbf{1}_{v_0=0,c_0=u_0} S_b(v_{\text{high}}) S_b((c+u)_{\text{high}}). \end{aligned}$$

Disjoint triples have distinct first-layer frequencies; shared-index terms are grouped by the frequency $\phi_0 - \phi_v$. Integrating their products uses the unique circuit paths and (6). This has a number of column sums of order n^3, and the integrals do not depend on k. The archive includes cross-checks against full spread RDMs, the all-layer certificate and the independent five-state transfer.

The sparse uniform comparison is still open: neither these finite tables nor the fixed-k asymptotics prove constants for $k \le \gamma n$ with fixed $\gamma < 1/2$. At half filling, the exact growing mode in Theorem 1 precludes a uniform upper constant. The neighborhood of half filling, where the contraction term in (1) becomes small, is a distinct remaining boundary problem.

Computation. All certificate and integration checks use exact rational arithmetic; no Lean formalization is claimed.

Use of AI tools. Haobo Ma and Wenlin Zhang used AI tools for search, double-checking and proofreading.

References

[1] I. Kerenidis, *Scalable Quantum Machine Learning: Trainability, Expressivity and Efficiency*, arXiv:2607.24014v2 [quant-ph], 2026. https://arxiv.org/abs/2607.24014v2.

[2] I. Kerenidis, correspondence with H. Ma and W. Zhang, 18 October 2026 (readout-recurrence identification and uniform-scalar question).